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3/28/24
CHE330

Extraction of D-Limonene from Orange Peels to Create a Predictive Model using DOE Analysis

In this DOE, I experimented with extracting D-limonene from the peels of oranges. Oranges are part of the citrus family, and are the fourth largest fruit produced in the world according to data from 2022. It has been a long interest of mine, as R-limonene has a sister molecule, S-limonene. S-limonene contributes to the smell of lemons, while R-limonene contributes to the smell of oranges. This ability to physically smell the chiral nature of molecules drew my attention, and its extraction gave me a perfect opportunity for this homework. I had three variables in my DOE. Firstly, the method of preparation of the peels: finely chopped by hand or pureed in a blender. Secondly, the time spent distilling: either one or two hours. Lastly, the heat applied, either 4 or 5 - corresponding to the burner level on our apartment stove. My findings show an interesting and clear trend; pureeing the peels had the largest impact on yield of R-limonene, followed by time left distilling and lastly the level of heat applied.

The distillery was set up in a “home chemist” manner. An oil bath in a glass pan provided an adequate solution to the problem of heating a round bottom flask with a flat stovetop. Around both I provided a skirt of aluminum foil to help retain heat for more efficient boiling. The vapor was then collected and condensed in a condensing tube, where it was collected. After collection, water was separated from the product and then the oil was measured. Since R-limonene is the dominant ingredient in orange peel oil (70-98%), I assumed my product to be pure for ease of calculation. In a real lab, the product could then be purified using ethanol or DMSO.

After gathering the data and making the model, it became clear that the largest factor affecting the yield was the preparation of the peel. When initially doing research for this experiment, I came across a paper discussing methods of optimizing extraction of limonene from orange peels, and they opted for cutting their orange peels into 2x5x3mm pieces, and I wanted to determine if that was the best method. Because of this, I chose to puree the peels or chop them finely, averaging a size of 2x2x2 mm. I theorized that pureeing the solution would allow for faster extraction of the R-limonene, and my results support my hypothesis. The other two factors, heat and time of distilling, had smaller effects on the yield. I believe this is because both heat settings were enough to boil the solution, giving it sufficient energy to vaporize. It is also clear that simply letting the solution boil for longer gives a higher yield, and I would be interested to see if one could model a curve of R-limonene extracted by time, I would hypothesize that the majority of the product occurs in the beginning, as limonene is more volatile than water.

Each trial consisted of 290 grams of pureed or chopped peel, with an initial 300 mL of water, followed by adequate water to ensure the solution did not dry out and burn. Using the data I acquired, I created a model to predict the yield of the limonene, which ended up accurately predicting the yield for my test run to within 3.4 mL of extract. I believe that some runs having large user errors could have affected the model’s accuracy. There was one run where the first 20 minutes the flask was not sealed, meaning all vapor was lost instead of collected. In addition, the low volumes of extract made it difficult to predict the volume of product, and larger batches or better measurement equipment would have allowed for more precise data.

In conclusion, I learned much about the expiration process, specifically how important some factors can be to the outcome. I also learned the best method of extracting limonene, which I will now promptly use to make orange essential oils for a variety of uses.

Raw Data

Factor	Trial	1	2	3	4	5	6	7	8
A	Sliced or Pureed	P	P	P	P	S	S	S	S
B	1 or 2 hours	1	2	2	1	1	2	2	1
C	4 or 5 heat	5	5	4	3.5	5	5	4	4
Result	mL obtained	5	6	4.5	4.5	2	1.5	0.2	0.5

Notes:
Pureed is +, Sliced is -
1 hour is -, 2 hours +
4 heat is -, 5 heat is +
Calculations

Ymodel	Value	Trial #	A	B	C	Actual	Model	Diff
B0	3.95	1	1	-1	1	5	7.9	2.9
B1	3.95	2	1	1	1	6	9.9	3.9
B2	0.05	3	1	1	-1	4.5	6.9	2.4
B3	1.2	4	1	-1	-1	4.5	6.9	2.4
B4	0.45	5	-1	-1	1	2	1.9	0.1
B5	-0.2	6	-1	1	1	1.5	0.9	0.6
B6	0.2	7	-1	1	-1	0.2	-1.7	1.9
B7	0.3	8	-1	-1	-1	0.5	-1.1	1.6
		TEST	1	0	1	5.5	8.9	3.4

Sources

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